



Connect And Protect: Networks and Network Security

Module 1

Network: A group of connected devices.

Local Area Network (LAN): A network that spans a small area like an office building, a school, or a home.

Wide Area Network (WAN): A network that spans a large geographic area like a city, state, or country.

Network tools:

Hub: A network device that broadcasts information to every device on the network.

Switch: A device that makes connection between specific devices on a network by sending and receiving data between them.

Router: A network device that connects multiple networks together.

Modem: A device that connects your router to the internet and brings internet access to the LAN.

Virtualization tools: pieces of software that perform network operations

Cloud computing: the practice of using remote servers, applications, and network services that are hosted on the internet instead of on local physical devices.

Cloud network: A collection of servers or computers that stores resources and data in remote data centers that can be accessed via the internet.

Data packet: A basic unit of information that travels from one device to another within a network.

Header Body Footer

Header contains IP address/MAC address/Protocol number

Body contains the message being transmitted to receiving device

Footer tells receiving device the packet is finished

Bandwidth: The amount of data a device receives every second.

Quantity of data / time in seconds = Bandwidth

Speed: The rate at which data packets are received or downloaded.

Packet sniffing: Practice of capturing and inspecting data packets across a network. (Wireshark)

Transmission Control Protocol (TCP): A internet communication protocol that allow two devices to form a connection and stream data.

Internet Protocol (IP): A set of standards used for routing and addressing data packets as they travel between devices on a network.

Port: A software-based location that organizes the sending and receiving of data between devices on a network.

Port divide network traffic into segments based on the service they will perform between two devices.

Port 25: Email

Port 443: HTTPS

Port 20: Large File transfers

TCP/IP Model: Framework used to visualize how data is organized and transmitted across the network.

4 Layers:

1. Network access layer:
 - a. Ethernet / Wireless LAN
2. Internet Layer
 - a. IP address / IP routing
3. Transport Layer
 - a. TCP / UDP
 - b. Error Control
4. Application Layer
 - a. HTTP / TLS / DNS

OSI Model:

7 Layers:

1. Physical Layer
 - a. Ethernet + wiring
 - b. Responsible for physical transmission of bits over hardware
2. Data Link
 - a. Mac Addressing / Frame Transmission
 - b. Responsible for communication on the same network
3. Network
 - a. IP address / IP routing
 - b. Responsible for logical addressing and routing packets between networks.
4. Transport
 - a. TCP / UDP
 - b. Responsible for end to end communication and reliability.
5. Session

- a. Keeping a session between a server and device active
 - b. Manages sessions (connection) between devices.
- 6. Presentation
 - a. Formats / Encryption / Compression
 - b. Responsible for data formatting, encryption, and compression
- 7. Application
 - a. HTTP / HTTPS
 - b. Closest layer to users and applications

Internet Protocol (IP) address: A unique string of characters that identifies the location of a device on the internet

IP version 4 (IPv4)

IP version 6 (IPv6)

Public IP

Private IP: Only on the same local network

Mac Address: A unique alphanumeric identifier that is assigned to each physical device on a network.

Module 2

Network Protocols: A set of rules used by two or more devices on a network to describe the order of delivery and the structure of the data.

Transmissions Control Protocol (TCP): An internet communications protocol that allows two devices to form a connection and stream data.

Address Resolution Protocol (ARP): A network protocol used to determine the MAC address of the next router or device on the path.

Hypertext Transfer Protocol Secure (HTTPS): A network protocol that provides a secure method of communication between clients and website servers.

Domain Name System (DNS): A network protocol that translate internet domain names into IP addresses.

Security Protocols:

HTTPS

SSL / TLS

Secure File Transfer Protocol (SFTP): Secure protocol to transfer files

Management Protocols:

Simple Network Management Protocol (SNMP): A network protocol for monitoring and managing devices on a network.

Internet Control Message Protocol (ICMP): Internet protocol used by devices to tell each other about data transmissions errors across the network.

IEEE 802.11 (WiFi): A set of standard that define communication for wireless LANs.

WiFi Protected Access (WPA): A wireless security protocol for devices to connect to the internet.

Firewall: A network security device that monitors traffic to and from your network.

Port filtering: A firewall function that blocks or allows certian port numbers to limit unwanted communication.

Hardware firewall: Physical device that inspects each data packet before it is allowed to enter the network.

Software firewall: Software that inspects each data packet before it is allowed to enter the network.

If software firewall installed on web server it will protect all devices on the network.

If software firewall isstalled on device it will protect only that device.

Cloud-based firewalls: software firewalls that are hosted by a cloud service provider.

Stateful: A class of firewall that keeps track of information passing through it proactively filters out threats.

Stateless: A class of firewall that operates based on predefined rules and does not keep track of information from data packets.

NGFWs

- Deep packet inspection

- Intrusion protection

- Threat intelligence

Virtual private networks (VPNs): A network security service that changes your public IP address and hides your virtual location so that you can keep data private when you are using a public network like the internet.

- VPN performs encapsulation on data

Encapsulation: A process performed by a VPN service that protects your data by wrapping sensitive data in other data packets.

Network segmentation: A security technique that divides the network into segments.

Security zone: A segment of a network that protects the internal network from the internet.

- Uncontrolled zone: Any network outside of the org's control

- Controlled zone: A subnet that protects the internal network from the uncontrolled zone.

- Demilitarized zone (DMZ): Webservers, proxy servers, DNS servers

- Internal Network

- Restricted zone: The most protected zone only with a certain level of access.

Internet Firewall DMZ Firewall Internal network firewall restricted zone

Subnetting: process of taking of large network and dividing it into smaller, organized groups call subnets.

Proxy server: A server that fullfills the requests of a client by forwarding them on to other servers.

Forward proxy server: regulates and restricts a person's access to the internet.

Reverse proxy server: Regulates and restricts the internets access to an internal server.

Module 3

Denial of service attack(DoS): An attack that targets a network or server and floods it with network traffic and overload it.

Distributed denial of service attack (DDoS): A type of DoS attack that uses multitple devices or servers in different locations to flood the target network with unwanted traffic.

Internet Control message protocol (ICMP): An internet protocol used by devices to tell each other about data transmission errors acroos network.

Network Level DoS attack:

SYN (Synchronize) flood attack: A type of DoS attack that simulates a TCP connection and floods a server with SYN packets.

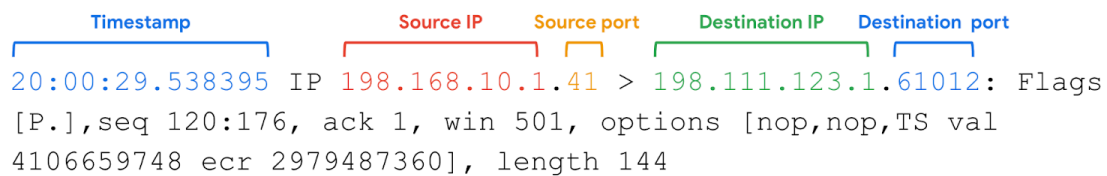
Internet Control message protocol (ICMP) flood: A type of DoS attack performed by an attacker repeatedly sending ICMP packets to a network server.

Ping of death: A type of DoS attack caused by when a hacker pings a system by sending it an oversized ICMP packet that is bigger than 64 KB.

tcpdump: Command-line protocol analyzer, provides a brief packet analysis and converts key information about network traffic into formats easily read by humans. It prints information about each packet directly into your terminal.

displays the source IP, destination IP, and the port numbers being used in the communication.

Interpreting out:



```
20:00:29.538395 IP 198.168.10.1.41 > 198.111.123.1.61012: Flags  
[P.], seq 120:176, ack 1, win 501, options [nop,nop,TS val  
4106659748 ecr 2979487360], length 144
```

Timestamp: hours, mins, seconds, fraction of seconds.

Source IP: the packets origin provided by its source IP.

Source port: this port number is where the packet orginiated.

Destination IP: The destination IP is where the packet is being transmitted to.

Destination port: This port number is where the packet is transmitted to.

Packets:

Header: MAC and IP address

Body: Data

Footer:

Packet sniffing: Practice of using software tools to observe data as it moves across networks.

Malicious packet sniffing:

Maclicious actors can insert themselves in the middle of an authorized connection between two devices

Passive packet sniffing: a type of attack where data packets are read in transit.

Active packet sniffing: a type of where data packets are manipulated in transit.

Prevent packet sniffing:

VPN tunnel: attackers might interfere with your traffic but they can't decode the data.

HTTPS: Uses SLS / TLS to encrypt data

Avoid using unprotected wifi

IP spoofing: A network attack performed when an attacker changes the source IP of a data packet to impersonate an authorized system and gain access to a network.

On-path attacks: An attack where a malicious actor places themselves in the middle of an authorized connection and intercepts or alters the data in transit.

Replay attack: A network attack performed when a malicious actor intercepts a data packet in transit and delays it or repeats it at another time.

Smurf attack: A network attack performed when an attacker sniffs an authorized user's IP address and floods it with packets.

Module 4

Attack surface: all the potential vulnerabilities that a threat actor could exploit.

Security hardening: the process of strengthening a system to reduce its vulnerability and attack surface.

Software updates

Patches

Device or application configuration changes.

Penetration test: A simulated attack that helps identify vulnerabilities in systems, networks, websites, applications, and processes.

Operating system (OS): the interface between computer hardware and the user.

Patch update: A software and operating system update that addresses security vulnerabilities within a program or product.

Baseline configuration (baseline image): A documented set of specifications within a system that is used as a basis for future builds, releases, and updates.

Network log analysis: The process of examining network logs to identify events of interest.

Security information and event management tool (SIEM): An application that collects and analyzes log data to monitor critical activities in an organization.

Network hardening:

Port filtering: A firewall function that blocks or allows certain port numbers to limit unwanted communication.

Cloud network: a collection of servers or computers that stores resources and data in remote data centers that can be accessed via the internet.